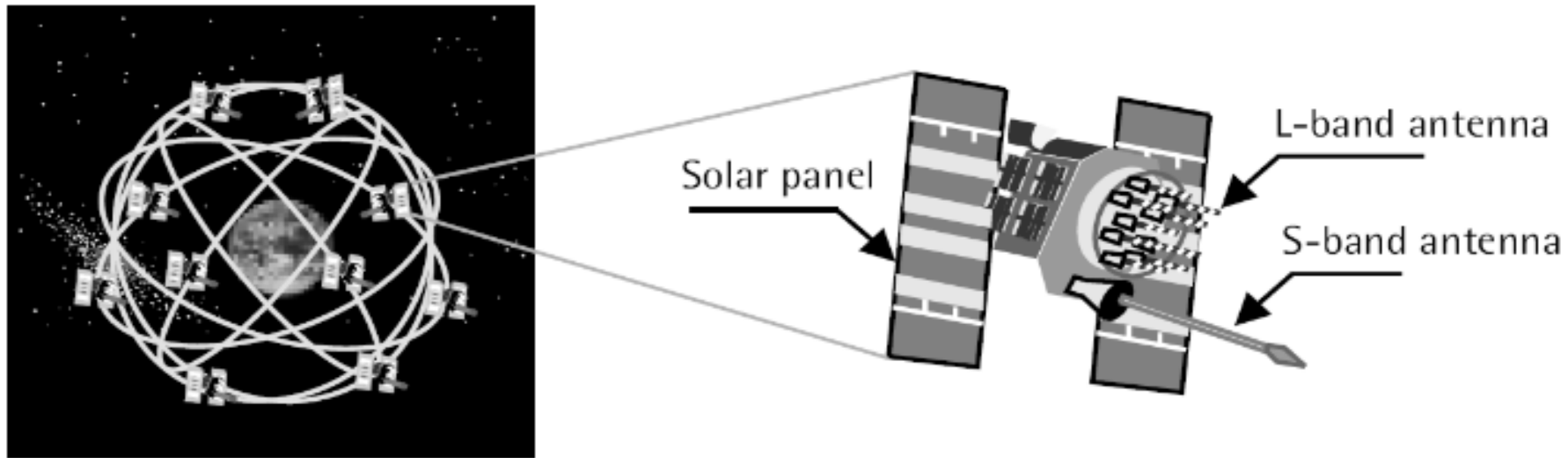


Unit-9.

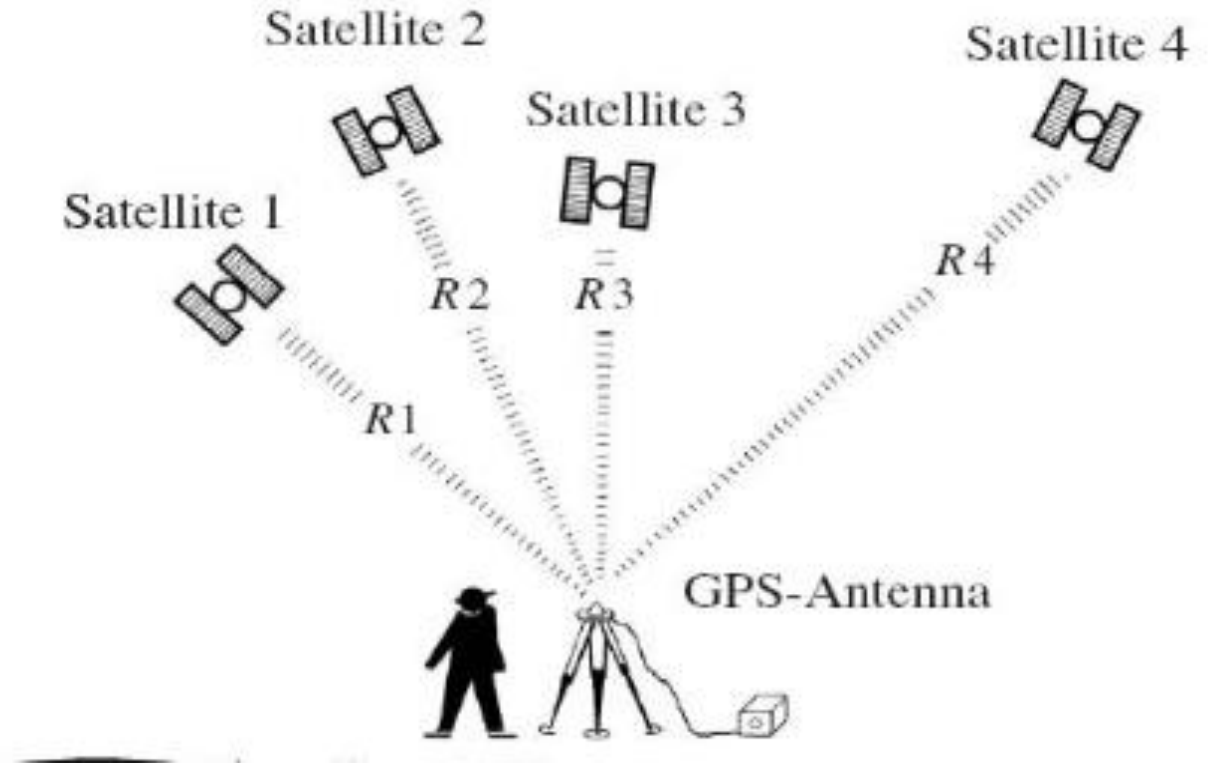
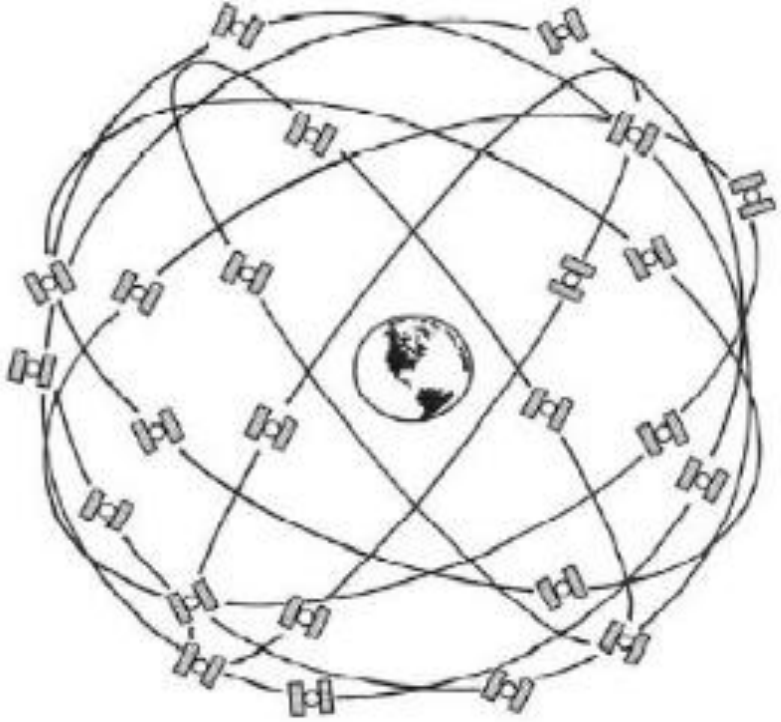
GPS - Global Positioning System (GPS)

Introduction:

GPS stands for Global Positioning System by which anyone can always obtain the location and time information anywhere in the world. It is a system of earth-orbiting satellites which provides a precise location on earth's surface in latitude/longitude coordinates. GPS is a powerful and widely used integrated tool (like in GNSS) that is used in Civil Engineering for collection of tabulated location data.



Orbits of GPS satellite



Characteristics of GPS satellite orbit:

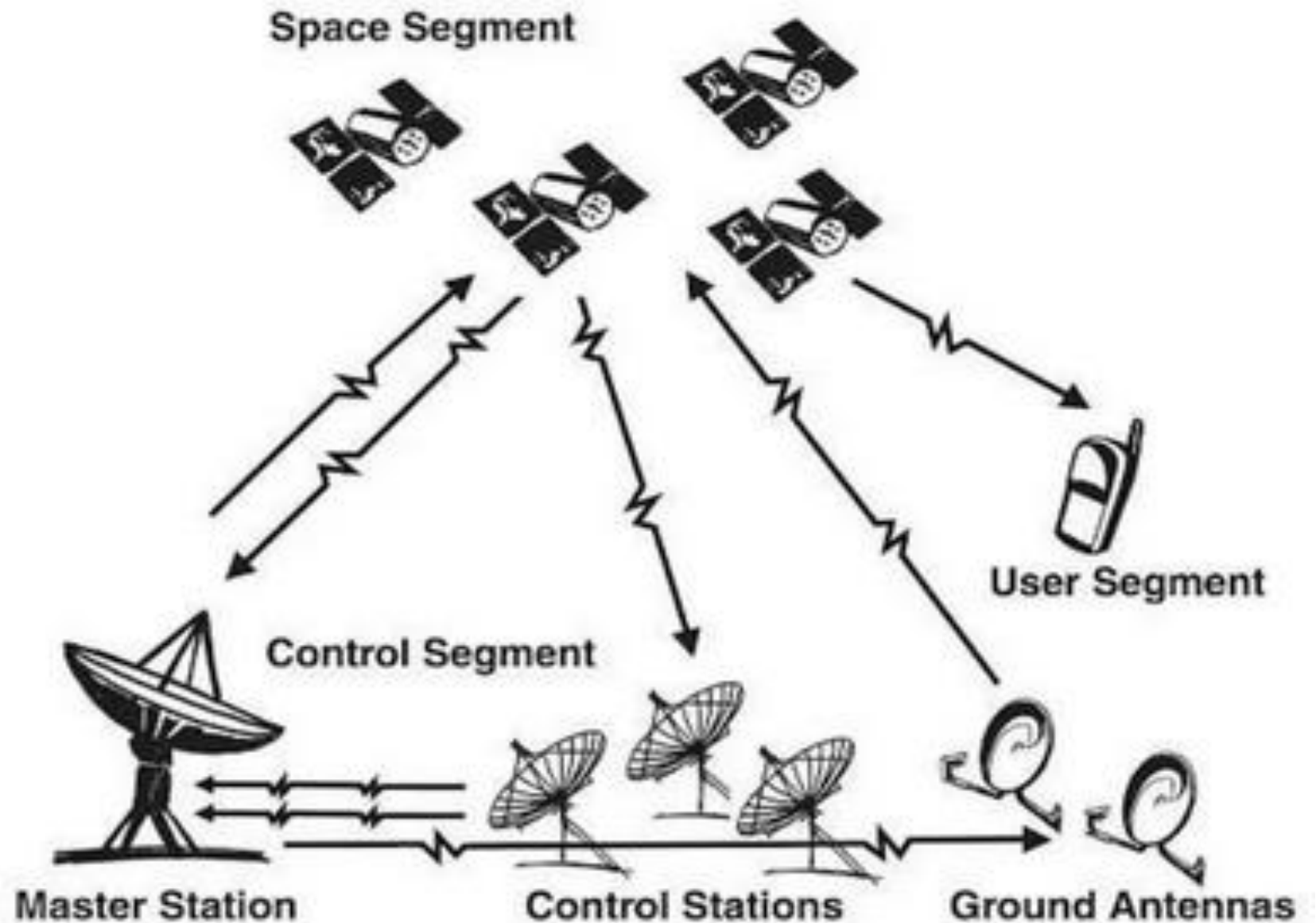
- Satellite revolves earth at a speed of 3.9 km/s, 11 h 58 min for complete revolution of earth.
- The height is 26560 km from the center of the earth. The height of the orbits above the earth is then about 20200 km.
- Satellites are arranged on 6 orbit, each at least 4 satellite where satellites can be arranged equidistantly.

The number and constellation of satellites guarantees that **the signals of at least four satellites can be received at any time all over the world.**

Components of GPS

Three component of GPS:

1. Space Segment.
2. Control Segment.
3. User Segment.



1. Space Segments –

A number of GPS fly in a circular orbit around the earth at an altitude of approximately 20,000 km (four GPS satellites per one orbit) with a period of 12 hours. At least 6 satellites are always within line of sight from any location and these satellites are powered by solar cells.

2. Control Segments –

Control Segment consists of the **Master station, monitor station, ground antenna**. Master station maintains optimum GPS constellation. Monitor station checks the exact altitude, position, speed of orbiting satellite. Ground antennas communicate with GPS satellite. It is responsible for the proper functioning of all the operations of GPS.

3. User Segments –

The user segment includes all military and civilian users.

- With a GPS receiver connected to a GPS antenna, a user can receive the GPS signals, which can be used to determine his or her position anywhere in the world.
- . At least 4 GPS satellites are required for calculating the exact position.

Working Principle of GPS

- GPS is based on the principle of resection i.e. Receiver's position (x,y,z) can be determined given the distances from objects whose positions are known.
- Analytically, from the known positions of three objects and their distance from unknown position, we get three-distance equation involving three known corresponding to the unknown position of the receiver.
- These three equations can be solved for three unknowns (x,y,z) .
- Theoretically, only three distances to three simultaneously tracked satellites are needed. R_1 , R_2 and R_3 in Fig: 1
- In this case the Receiver would be located at the intersection of three spheres each has a radius of one receiver-satellite distance. Fig: 2

Resection

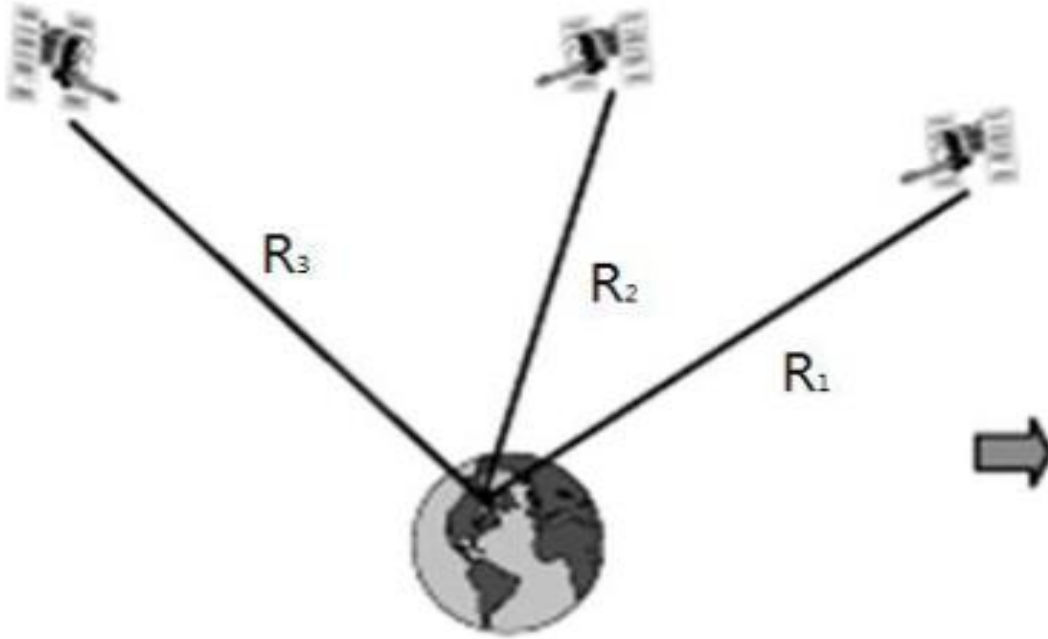


Fig: 1 **2D position determination with 3 satellites**

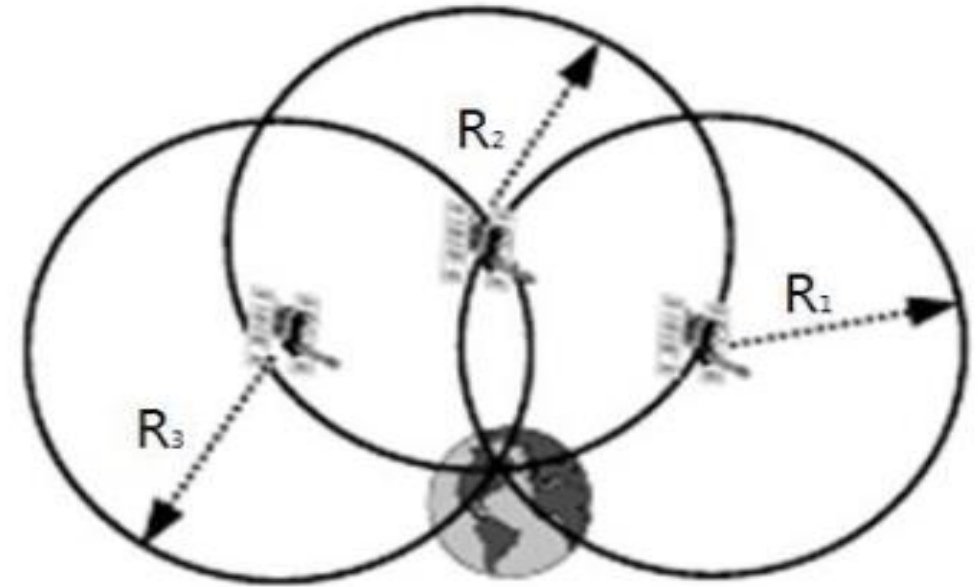


Fig: 2

- Then the intersection of three distances from satellite to the receiver would be the position of receiver.
- Satellites broadcast their position as a function of time in WGS-84 Coordinates system.
- Therefore, the position of receiver is determined in WGS-84 coordinate system, which later converted into local coordinate system.

How to calculate Distance from Satellite to Receiver?

- In GPS, the satellites, which broadcast their positions, are the objects at known locations.
- The distance between the receiver and a satellite is measured in terms of transit time of the signal multiplied by velocity of light.

i.e. Distance = Velocity x Time

- The velocity is the velocity of radio signal broadcasted by the satellite at the speed of light (3×10^{10} cm/sec).
- There is the time taken for the radio signal to travel from the satellite to the GPS receiver.
- This is little harder to calculate, since we need to know when the radio signals left the satellite and when it reached the receiver.
- Therefore, the travel time is fourth unknown to be estimated in addition to the three coordinates of receiver's position.
- A user, therefore, needs a minimum of four satellites in view to estimate his four-dimensional position: three coordinates for spatial position plus time bias.

Application of GPS in Civil Engineering:

Global Positioning System (GPS) is a highly accurate navigation system that's used in Civil engineering survey for a variety of purposes, including:

- **Measuring distances:** GPS can accurately measure distances between points in a given area.
- **Plotting boundaries:** GPS is used to plot the boundaries of properties and other areas of interest.
- **Topographic mapping:** GPS is useful for topographic mapping
- **Engineering design:** GPS surveying can provide accurate data to help civil engineers draft proposals.
- **Construction layout:** GPS surveying can be used to ensure structures are placed in optimal positions.
- **Surveying coasts and waterways:** GPS is especially useful in surveying coasts and waterways because there are few land-based reference points.
- **Hydrographic surveys:** Bridge builders and offshore oil rigs use GPS for accurate hydrographic surveys.

Here are some other uses of GPS: Navigation, Tracking, Mapping, and Timing.

Remote sensing and GIS

Introduction to Remote sensing:

The science (and art) of acquiring information about an object, without entering in contact with it, by sensing and recording reflected or emitted energy and processing, analyzing, and applying that information.

Thus, remote sensing is the process of inferring surface parameters from measurements of the electromagnetic radiation (EMR) from the Earth's surface. This EMR can either be reflected or emitted from the Earth's surface.

In other words, remote sensing is detecting and measuring electromagnetic (EM) energy emanating or reflected from distant objects made of various materials, so that we can identify and categorize these objects by class or type, substance and spatial distribution.

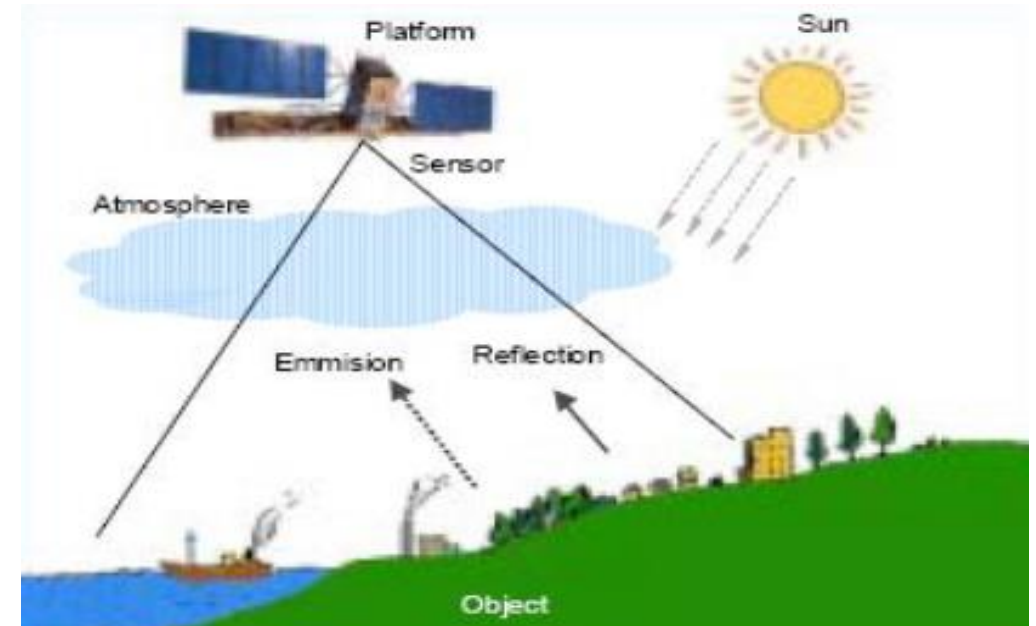


Fig. Shows Schematic representation of remote sensing technique.

Components of Remote Sensing

1. Platform:

A Platform is defined as the carrier for remote sensing sensors. There are three major remote sensing platforms:

- a) Ground-level platform: (towers and cranes),
- b) Aerial platforms: (Helicopters, low altitude aircraft, high altitude aircraft), and
- c) Space borne platforms: (space shuttles, polar-orbiting satellites, and geostationary satellites).



Fig. 1: Ground-Based Platform

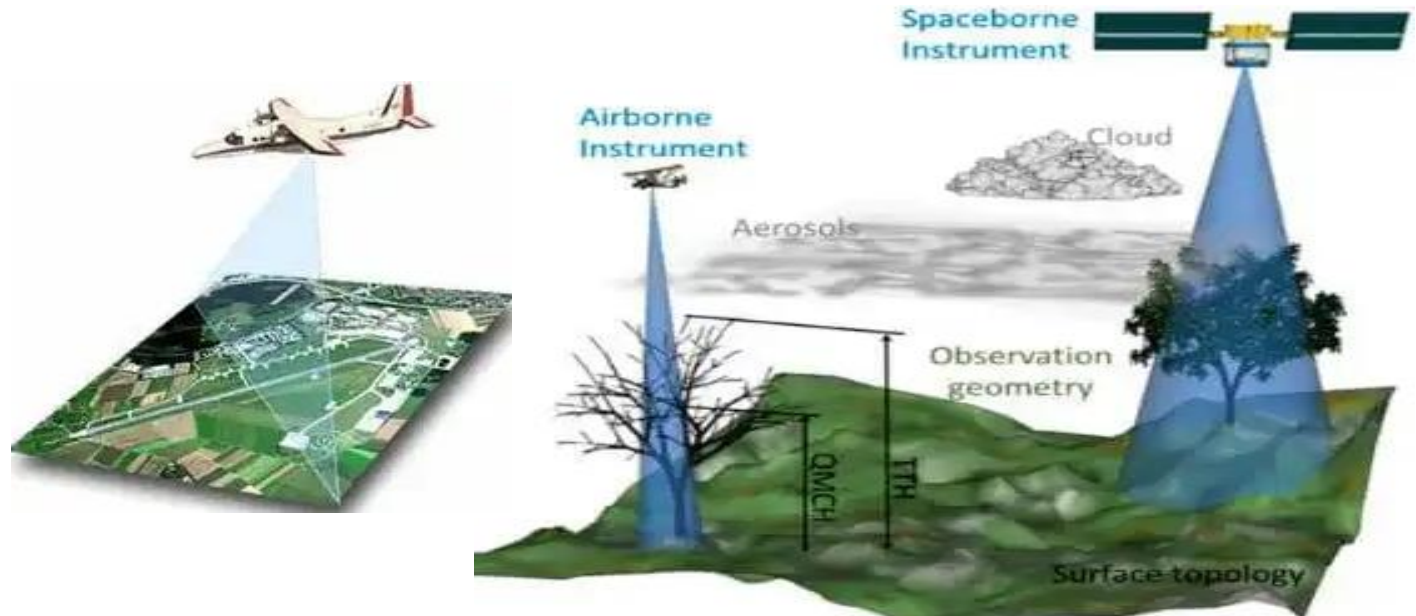


Fig. 2: Airborne and space based Platform

2. Sensors:

It is a **device that receives electromagnetic radiations** and converts it into a signal that can be recorded and displayed as either numerical data or an image.

Types of Remote Sensing

Sensors: Sensors are of two kinds-passive and active (in term of energy source).

The classification of remote sensing mainly shows dependency on the type of source that produces the electromagnetic energy. On the basis of the source of electromagnetic radiation production, remote sensing is classified as active and passive.

Passive Remote Sensing:

The type of remote sensing that makes use of a natural source of energy i.e., the sun in order to produce electromagnetic radiation and perform remote sensing is called passive remote sensing. Generally, solar energy is used in various remote sensing applications i.e., the passive mode of operation. The solar energy gets reflected by the targets at a specific wavelength is sensed by the sensors.

The strength of the signal received in this case interacts with the atmosphere in an efficient manner and thus no significant loss in signal strength is noticed. This is similar to an analogy like taking an image from an ordinary camera.

Active Remote Sensing:

In this type of remote sensing energy is produced from a natural source that is present on the platform and is sent towards the target. The sensors present on the platform collect the reflected energy from the target. An example analogy of active remote sensing is capturing pictures from a device with built-in flash.

PASSIVE VS ACTIVE REMOTE SENSING

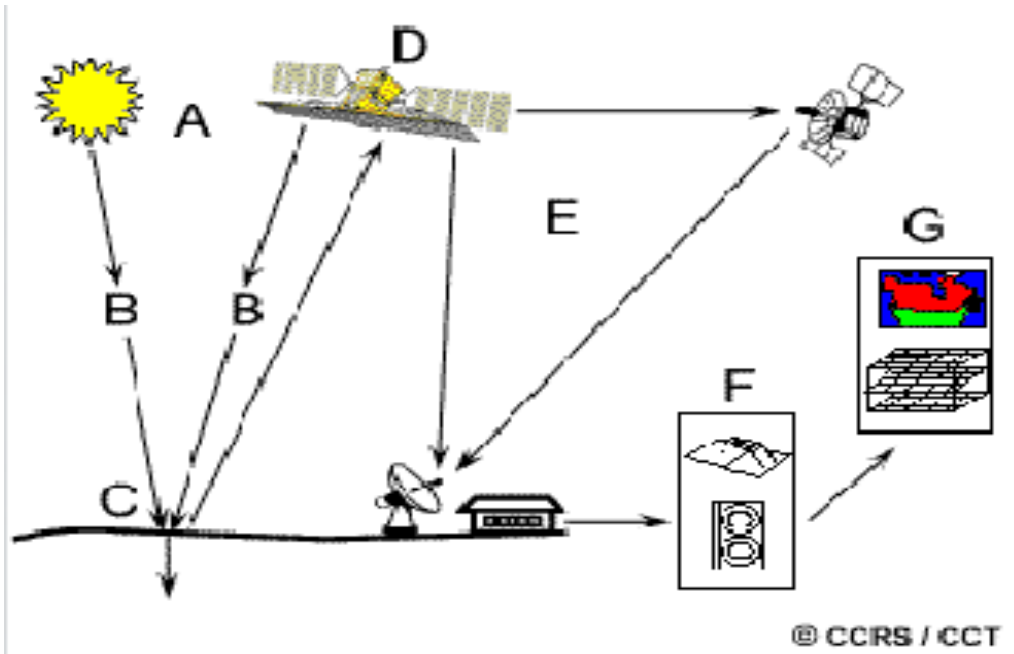
Passive Remote Sensing

- ✓ Does not employ any external source of energy.
- ✓ Measure either reflected radiation from sun (can be operated only during daytime) or the emitted radiation from the surface (day/ night operation).
- ✓ Suffers from variable illumination conditions of sun and influence of atmospheric condition.

Active Remote Sensing

- ✓ Has its own source of energy.
- ✓ Active sensors emit a controlled beam of energy to the surface and measure the amount of energy reflected back to the sensor.
- ✓ Controlled illumination signal.
- ✓ Day/night operation.

Process and Stages of Ideal Remote Sensing:



1. Energy Source or Illumination (A)- an energy source illuminates or provides electromagnetic energy to the target of interest.

2. Radiation and the Atmosphere (B)- as the energy travels from its source to the target, it will come in contact with and interact with the atmosphere it passes through.

3. Interaction with the Target (C)- interacts with the target depending on the properties of both the target and the radiation.

4. Recording of Energy by the Sensor (D)- after the energy has been scattered by, or emitted from the target, we require a sensor to collect and record the electromagnetic radiation.

5. Transmission, Reception, and Processing (E)- the energy recorded by the sensor has to be transmitted, often in electronic form, to a receiving and processing station where the data are processed into an image (hardcopy and/or digital).

6. Interpretation and Analysis (F)- the processed image is interpreted, visually and/or digitally or electronically, to extract information about the target which was illuminated for different purpose or application.

7. Application (G) - Apply the information we have been able to extract from the imagery about the target in order to better understand it, reveal some new information, or assist in solving a particular problem.

Application of remote sensing

Remote sensing is used in civil engineering for a variety of applications, including:

Urban planning

Remote sensing provides information on land use, population density, and infrastructure distribution, which can help with urban planning and development.

Surveying and mapping

Remote sensing technologies like satellite photography and LiDAR (Light Detection and Ranging) can be used to create maps and perform surveys. These technologies can be especially useful for difficult-to-reach terrain.

Environmental monitoring

Remote sensing and Geographic Information Systems (GIS) can be used to map and analyze environmental parameters like land cover, land use, vegetation health, and water quality. This information can be used to monitor environmental change and support sustainable development.

Geophysics

Remote sensing uses electromagnetic radiation to collect data from afar, providing insights into Earth's features and processes.

Remote sensing is used in many other fields, including agriculture, disaster management, climate monitoring, urban planning, weather forecasting, and mining.

Advantages

1. It helps in collecting data of large areas.
2. Easy way of collecting data.
3. Remote locations are accessible easily.
4. It is inexpensive in comparison to using a team of surveyors for observing or collecting information on objects or regions.
5. From collected data, maps can be rapidly produced.

Disadvantages

1. Skilled users are needed for interpretation.
2. A large amount of data sometimes create confusion.
3. Cross verification of data is required.
4. As sensors are mobile thus their motion sometimes causes distortions in the image.

System of hardware, software, data and organizational structure for capturing, storing, checking, integrating, manipulating, analyzing and displaying data related to positions on the Earth's surface.

Components of a GIS

A working GIS integrates five key components: hardware, software, data, people, and methods.

1. Hardware:

Hardware is the computer on which a GIS operates. Today, GIS software runs on a wide range of hardware types, from centralized computer servers to desktop computers used in stand-alone or networked configurations.

2. Software:

GIS software provides the functions and tools needed to store, analyze, and display geographic information.

Key software components are:

- Tools for the input and manipulation of geographic information.
- A database management system (DBMS).
- Tools that support geographic query, analysis, and visualization.
- A graphical user interface (GUI) for easy access to tools

3. Data: Possibly the most important component of a GIS is the data. Geographic data and related tabular data can be collected in-house or purchased from a commercial data provider. A GIS will integrate spatial data with other data resources and can even use a DBMS, used by most organizations to organize and maintain their data, to manage spatial data.



- 4. People:** GIS technology is of limited value without the people who manage the system and develop plans for applying it to real-world problems. GIS users range from technical specialists who design and maintain the system to those who use it to help them perform their everyday work.
- 5. Methods:** A successful GIS operates according to a well-designed plan and business rules, which are the models and operating practices unique to each organization.

Application of GIS in Civil Engineering:

- Ø Landslide hazard zonation
- Ø Site selection for groundwater potential zones
- Ø Hurricane wind Analysis
- Ø Natural Hazard Assessment
- Ø Landfill Site Selection
- Ø Pollution Monitoring
- Ø Watershed Analysis
- Ø Site selection for solid waste disposal
- Ø Earthquake probability Analysis
- Ø Transportation analysis
- Ø Target Site Selection
- Ø Mineral mapping
- Ø Urban Development

THANK YOU